Topographical Orthogonal Generative Theory (TOGT)

Volume Two: Renormalization, Topological Invariants,
and the Crystalline Spine

Structural Outline

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Abstract

This document is a structural outline for Volume Two of TOGT. It organises the four open research directions identified in Volume One into a coherent sequence of sections, specifying dependencies, provable targets, and conjectural directions. No proofs or mathematical claims are made here; the purpose is architectural scaffolding only.

Two typographic markers are used throughout: **[Provable target]** marks goals for which a proof strategy is in view and the mathematical prerequisites are in place; **[Conjectural]** marks directions that require new definitions, external formalisation, or unresolved foundational questions before a proof can be attempted.

Contents

Part I
Function-Space Foundations for the Renormalization Operator

1 Unimodal Map Spaces and Their Norms

Description

This section establishes the precise function space in which the renormalization operator $\mathcal{R}$ will be defined. The two natural candidates — analytic unimodal maps on a complex neighbourhood of $[-1, 1]$, and C^3 unimodal maps with negative Schwarzian derivative — are compared, and the relevant Banach or Fréchet norms are recorded. The section closes by fixing one setting as the working hypothesis for the rest of Volume Two, with a justification based on the regularity of the Euler-discretized gradient flow $f_{\lambda,h}$.

Status

[Provable target] The definitions and norm comparisons are standard material from the Collet–Eckmann and Sullivan–McMullen literature and require no new mathematics; they are imported and specialised to the TOGT setting.

Depends on (V1): Proposition 3.2 (V1, §3.3): the Euler discretization $f_{\lambda,h}$ is a C^∞ unimodal map with cubic critical point, hence lies in both candidate spaces for h small enough.

2 The Renormalization Operator as a Map on Function Space

Description

Having fixed the function space, this section defines $\mathcal{R}$ as a rigorous operator

$$\mathcal{R} : \mathcal{U} \to \mathcal{U}, \qquad \mathcal{R}(f) = \alpha^{-1} \circ f^2 \circ \alpha,$$

where $\mathcal{U}$ is the chosen space of unimodal maps and α is the Feigenbaum rescaling. Domain invariance — the question of whether $\mathcal{R}(f)$ remains unimodal when f is — is verified. Basic continuity and compactness properties are established.

Status

[Provable target] Domain invariance and continuity follow from the classical theory (Collet–Eckmann [1]); the contribution here is specialising those results to the TOGT operator family and verifying that $f_{\lambda,h}$ lies in the domain.

Depends on (V1): Section 1 of this volume (function-space choice); Proposition 3.2 (V1): regularity of $f_{\lambda,h}$.

3 Relation Between the Gradient Flow and Its Discretization

Description

The gradient flow $\boldsymbol{g}_\lambda$ itself is not a unimodal map and does not lie in $\mathcal{U}$; Volume One states this explicitly (§3.3). This section makes the relationship precise: it characterises the map $h \mapsto f_{\lambda,h}$ and studies whether the renormalization orbit of $f_{\lambda,h}$ converges as $h \to 0$ to a well-defined limit that can be attributed to the continuous flow. This is the key step needed to connect $\mathcal{R}$ to the TOGT pipeline proper.

Status

[Conjectural] The $h \to 0$ limit of the renormalization orbit is not covered by existing results. A rigorous treatment requires either a continuous-time renormalization theory (non-standard) or a controlled discretisation argument. This is the primary new mathematical challenge of Part I.

Depends on (V1): Definition of $\boldsymbol{g}_\lambda$ (V1, §3.1); Proposition 3.2 (V1): domain membership of $f_{\lambda,h}$; Section 2 of this volume: $\mathcal{R}$ well-defined on $\mathcal{U}$.

Part II
Dynamics of the Renormalization Operator

4 The Feigenbaum Fixed Point: Existence and Uniqueness

Description

This section imports the Feigenbaum–Coullet–Tresser fixed-point theorem (as proved analytically by Lanford, and later by Lyubich) into the TOGT setting. The fixed point $\boldsymbol{g}_\infty \in \mathcal{U}$ of $\mathcal{R}$ is identified, and its basic properties — symmetry, analyticity, and the values of the universal constants

$\alpha \approx 2.5029$ and $\delta \approx 4.6692$ — are recorded. The section does not re-prove the fixed-point theorem; it establishes that the TOGT operator $f_{\lambda,h}$ lies in the basin of attraction.

Status

[Provable target] The fixed-point theorem is established in the literature; the work here is verifying the basin condition for $f_{\lambda,h}$, which follows from Proposition 3.2 (V1) and the universality class membership.

Depends on (V1): Proposition 3.2 (V1): $f_{\lambda,h}$ lies in the period-doubling universality class; Section 2 of this volume: $\mathcal{R}$ well-defined.

5 Hyperbolicity of the Renormalization Horseshoe

Description

The renormalization operator $\mathcal{R}$ has a hyperbolic fixed point $\boldsymbol{g}_\infty$ with a one-dimensional unstable manifold (corresponding to the quadratic family parameter) and an infinite-codimensional stable manifold (the space of infinitely renormalizable maps of the same combinatorial type). This section states the Avila–Lyubich hyperbolicity theorem precisely and identifies its implications for the TOGT pipeline: in particular, which parameters (λ, h) yield infinitely renormalizable maps.

Status

[Provable target] Hyperbolicity is established by Avila–Lyubich [4]; the contribution is computing or bounding the set of (λ, h) for which $f_{\lambda,h}$ is infinitely renormalizable.

Depends on (V1): Section 4 of this volume: fixed-point identification; the contraction and pitchfork results (V1, §§5–6): they constrain the parameter range of interest.

6 Stable Manifolds and the Basin of the Universal Fixed Point

Description

The stable manifold of $\boldsymbol{g}_\infty$ under $\mathcal{R}$ consists of all infinitely renormalizable maps with the same combinatorial type (period doubling). This section characterises this manifold within $\mathcal{U}$, identifies which maps $f_{\lambda,h}$ lie on it, and studies the geometry of the foliation of $\mathcal{U}$ by stable leaves. This is the renormalization-theoretic underpinning of universality: two maps on the same leaf are asymptotically indistinguishable under $\mathcal{R}^n$.

Status

[Conjectural] The foliation theorem is known in the analytic category (Lyubich [2]); its extension to the specific family $\{f_{\lambda,h}\}$ and the identification of which leaf this family belongs to require explicit computation.

Depends on (V1): Section 5 of this volume: hyperbolicity; Section 3 of this volume: $h \to 0$ limit.

7 Scale-Invariance of the TOGT Pipeline Under Renormalization

Description

This section assembles the results of Part II into a statement about the TOGT pipeline: under iterated renormalization, the operator $\boldsymbol{g}_\lambda$ (via its discretization) converges to the universal fixed point $\boldsymbol{g}_\infty$, and the large-scale structure of the pipeline $S(L) \to \Phi \to \boldsymbol{g} \to \mathcal{K}(L) \to$ Space becomes independent of λ. This is the TOGT analogue of universality in the classical sense. The extended pipeline $S(L) \to \Phi \to \boldsymbol{g}_\lambda \xrightarrow{\mathcal{R}^n} \boldsymbol{g}_\infty \to \mathcal{K}(L) \to$ Space proposed in Volume One (§3.3) is given its first rigorous content here.

Status

[Conjectural] Full rigour requires resolving the $h \to 0$ problem of Section 3. The result is provable conditional on that resolution.

Depends on (V1): The full extended pipeline (V1, §3.3); Sections 4–6 of this volume: fixed point, hyperbolicity, stable manifolds.

Part III
The Topological Invariant 33

8 Candidate Definitions of the Topological Invariant

Description

The scalar $m(\mathcal{M}(g^6(X))) = 33$ is stated as a conjecture in Volume One (§13) but the readout m is not yet precisely defined. This section surveys three candidate definitions: (i) the sum of Betti numbers of the sublevel-set filtration of the iterated state; (ii) the total persistence (sum of lifetimes in the persistence diagram); and (iii) a Morse-theoretic count of critical points of a smooth functional associated to $g^6(X)$. Each candidate is evaluated against the reproducibility protocol of Volume One (§13.2) and the robustness data (Table 1, V1).

Status

[Provable target] The definitions themselves require no new mathematics; the contribution is selecting the one that matches the experimental data and verifying that the selection is well-posed.

Depends on (V1): The Betti-sum protocol and robustness table (V1, §§13.2–13.3); the honest-status remark (V1, §13): m requires a precise definition before the conjecture is fully stated.

9 The Morse Complex of Iterated States

Description

Given a chosen definition of m, this section constructs the Morse complex $\mathcal{M}(g^k(X))$ for the iterated states produced by the TOGT pipeline. It studies how the complex evolves under iteration — how many critical points appear or merge at each step — and whether there is a stabilisation phenomenon consistent with $m = 33$ at $k = 33$. The TTK pipeline described in Volume One (§13.2) is the computational model; this section provides the mathematical framework it implicitly uses.

Status

[Conjectural] The stabilisation at $m = 33$ is the empirical claim of Volume One; whether it holds rigorously for the specific pipeline operators ($\mathcal{C}omp, \mathcal{K}onst, \mathcal{F}old, \mathcal{U}nfold$) and any admissible X is open. Proving it would require control over the topology of the image of g^k for all admissible X.

Depends on (V1): The four-operator pipeline (V1, §2); the TTK protocol (V1, §13.2); Section 8 of this volume: candidate definition of m.

10 The Algebraic–Topological Separation, Revisited

Description

The General Separation Theorem (V1, Theorem 12.2) proves that 33 cannot arise as an algebraic trace $\mathrm{Tr}(M_n^6)$ for any stable $n < 33$ representation. This section revisits that result in light of the topological invariant defined in Section 8: it asks whether $m = 33$ is genuinely topological in the sense of being outside every algebraic invariant of the linearized pipeline, and whether the separation is sharp (i.e., whether there exists an $n = 33$ representation achieving the bound). This sharpens the bridge between the two tiers of the theory.

Status

[Provable target] (sharpness of the bound) The bound $|\mathrm{Tr}(M^6)| \leq n$ is attained when all eigenvalues equal 1 (V1, Corollary 11.1); the $n = 33$ case is therefore already provable from Volume One. The genuine topological content of $m = 33$ is conjectural pending Section 9.

Depends on (V1): Theorem 12.2 and Corollary 11.1 (V1): algebraic trace bound and separation; Section 9 of this volume: the Morse complex.

11 Behaviour of the Topological Invariant Under Renormalization

Description

This section addresses open problem (ii) from Volume One (§3.3): whether the topological invariant $m(\mathcal{M}(g^6(X))) = 33$ is preserved, transformed, or destroyed under $\mathcal{R}^n$. The heuristic computation of Volume One (§3.4) identified the degree-doubling assumption as the critical gap; this section formulates that assumption precisely in terms of the chosen definition of m (Section 8) and states what a proof would require.

Status

[Conjectural] No proof strategy is currently available. The section records the precise mathematical content of the open problem and the conditions under which invariance, covariance, or transformation could be established.

Depends on (V1): Section 8 of this volume: definition of m; Sections 4–7 of this volume: renormalization dynamics; The heuristic argument (V1, §3.4): identifies the degree-doubling gap.

Part IV
The g^{64} Crystalline Spine

12 The $n = 64$ Operator Algebra

Description

The Kether Orthogon conjecture (V1, Conjecture 14.1) posits that g^{64} achieves a terminal crystalline fixed point with topological readout 33. A prerequisite is the 64-dimensional operator algebra: the matrix M_{64} satisfying $\rho(M_{64}) = 1$, one contractive block, and one expansive direction. This section constructs M_{64} as a natural extension of the canonical matrix M_c (V1, §11.1), verifies that the algebraic trace bound gives $|\operatorname{Tr}(M_{64}^{64})| \leq 64$, and identifies which eigenvalue configurations would make $33 < 64$ a consistent reading across both the algebraic and topological layers.

Status

[Provable target] The construction of M_{64} and the trace bound follow directly from Theorem 12.1 (V1) with $n = 64$; no new mathematics is required beyond specifying the block structure.

Depends on (V1): Theorem 12.1 (V1): algebraic trace bound for general n; the canonical matrix M_c (V1, §11.1); Corollary 11.1 (V1): attainability of the bound.

13 Is g^{64} a Fixed Point of $\mathcal{R}$?

Description

The Kether Orthogon conjecture requires g^{64} to be an attractor or fixed point of the renormalization flow. This section formulates that question precisely: does the renormalization orbit $\mathcal{R}^n(f_{\lambda,h})$ converge to g^{64} (or to a map whose 64-fold iterate has the conjectured topological structure), or does it converge to g_∞ (the Feigenbaum fixed point) independently of iteration count? The two conjectures are shown to be compatible only under specific conditions on the relationship between g_∞ and g^{64}.

Status

[Conjectural] The relationship between g_∞ (Feigenbaum fixed point) and g^{64} (crystalline spine) is not currently understood. This section records the compatibility conditions as a precise open problem.

Depends on (V1): Conjecture 14.1 (V1): Kether Orthogon; Sections 4–7 of this volume: renormalization dynamics; Section 12 of this volume: $n = 64$ algebra.

14 The Orthogonality Condition

Description

The Kether Orthogon conjecture also posits an orthogonality condition $\langle \mathcal{M}(g^{64}(X)), g^{64}(X)\rangle_{\mathrm{TTK}} = 0$ (V1, §14). Before this can be studied, the TTK inner product $\langle \cdot, \cdot \rangle_{\mathrm{TTK}}$ on persistence modules must be defined. This section surveys candidate inner products on the space of persistence diagrams (e.g., the Wasserstein-based pairing, the kernel inner product of Reininghaus et al.), evaluates each against the orthogonality condition, and selects a working definition.

Status

[Conjectural] The TTK inner product is explicitly listed as undefined in Volume One (§14); this section is the first step toward resolving that gap. No orthogonality claim is made until a definition is fixed.

Depends on (V1): Conjecture 14.1 (V1): orthogonality condition; Section 9 of this volume: Morse complex construction; Section 10 of this volume: topological invariant under renormalization.

15 The g^{64} Spine as Terminal Attractor: Synthesis

Description

This closing section assembles the results of Part IV into a unified statement — conditional on the conjectures of Sections 13 and 14 — about the terminal behaviour of the TOGT pipeline. It states what a complete proof of the Kether Orthogon conjecture would require, which components are now within reach (the trace bound, the $n = 64$ algebra), and which remain genuinely open (the orthogonality condition, the relationship to g_∞). It also identifies the Lean 4 formalisation targets that would close the remaining `sorry`s in the Volume One nucleus.

Status

[Conjectural] This section is explicitly synthetic; no new results are claimed. It serves as a research agenda for the parts of Volume Two not yet resolved.

Depends on (V1): All proved results in Volume One; Sections 12–14 of this volume; The Lean nucleus (V1, §7): identifies the two scoped `sorry`s that formalisation of this section would discharge.

Summary of Dependencies and Status

The table below summarises each section of Volume Two, its status, and its primary dependency in Volume One.

§	Title (abbreviated)	Status	Primary V1 dependency
1	Unimodal map spaces	Provable	Prop. 3.2
2	$\mathcal{R}$ on function space	Provable	Prop. 3.2, §3.3
3	Flow vs. discretization	Conjectural	Prop. 3.2, §3.1
4	Feigenbaum fixed point	Provable	Prop. 3.2
5	Hyperbolicity	Provable	§§5–6
6	Stable manifolds	Conjectural	§§5–6, V2 §5
7	Scale-invariance	Conjectural	§3.3, V2 §§4–6
8	Candidate definitions of m	Provable	§§13.2–13.3
9	Morse complex	Conjectural	§2, §13.2
10	Separation revisited	Provable	Thm. 12.2, Cor. 11.1
11	m under renormalization	Conjectural	§3.4, V2 §§4–7
12	$n = 64$ operator algebra	Provable	Thm. 12.1
13	g^{64} as fixed point	Conjectural	Conj. 14.1
14	Orthogonality condition	Conjectural	Conj. 14.1
15	Terminal attractor synthesis	Conjectural	All of V1

Of the fifteen sections, six have clear proof strategies using results already established in Volume One (§§1, 2, 4, 5, 8, 10, 12). The remaining nine depend on new definitions, external formalisation, or unresolved foundational questions. The provable sections should be completed first; they form the scaffold on which the conjectural sections can be attempted.

References

[1] P. Collet and J.-P. Eckmann. *Iterated Maps on the Interval as Dynamical Systems.* Birkhäuser, Boston, 1980.

[2] M. Lyubich. Feigenbaum–Coullet–Tresser universality and Milnor's hairiness conjecture. *Ann. of Math.* **149** (1999), 319–420.

[3] A. Avila and G. Forni. Weak mixing for interval exchange transformations and translation flows. *Ann. of Math.* **165** (2006), 637–664.

[4] A. Avila and M. Lyubich. The full renormalization horseshoe for unimodal maps of higher degree: exponential contraction along hybrid classes. *Publ. Math. Inst. Hautes Études Sci.* **114** (2011), 171–223.